Neural Networks for Data Science

Master's Degree in Data Science

Lecture 0: About the course

Lecturer: S. Scardapane

SAPIENZA
UNIVERSITÀ DI ROMA

- **Master's Degree in Data Science**, code 10589627, 2nd year, optional group D.
- **Timetable**: TBD
- **Office hours**: by appointment, remotely or in-person (Via Eudossiana 18, DIET Department, 1st floor, room 122).

Official course website:
`https://www.sscardapane.it/teaching/nnds-2026/`.

Register to the **Google Classroom** from the website for all updates (mandatory).

1. **Sessions 1-2**: TBD
2. **Sessions 3-4**: TBD
3. **Session 5**: TBD
4. **Session E1**: TBD (**reserved**, see regulations).
5. **Session E2**: TBD (**reserved**, see regulations).

1. One **end-of-term** project (5 points).
2. One **oral examination** on the program (25 points).

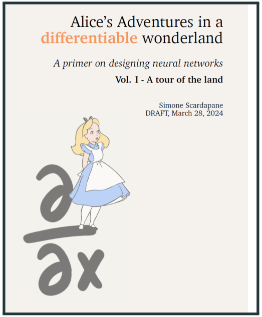
The EoT project can be sent before *any* exam date. The mark can be kept during the academic year, irrespective of the oral. *Lode* will be given only to exceptional (top 5%) projects and orals.

- Broad historical overview of the last 15 years.
- Fundamental tools underlying neural networks: optimization, gradient descent, automatic differentiation.
- Basic blocks to build modern neural networks (convolution, attention, normalization, ...).
- Proficiency in a real-world deep learning library (JAX, PyTorch).
- Capability of navigating the current literature and ecosystem in autonomy.

1. **Historical overview**.
2. **Preliminaries** (tensors, linear algebra, optimization).
3. **Supervised learning** as numerical optimization.
4. Linear models and **fully-connected** models.
5. **Convolutional models** (for sequential, spatial, and temporal data).
6. Blocks to train **deeper models** (dropout, batch normalization, ...).
7. **Attention** models for sets.
8. **LLM** building blocks: autoregression, tokenization, pre- and post-training.
9. **Graph** models (e.g., graph convolutional networks).
10. Optional topics depending on time and material.

1. Several practical lectures with JAX and PyTorch (hands-on coding from scratch).
2. When possible, a showcase of other libraries (e.g., HuggingFace Datasets).
3. Optional topics depending on time and material.

Slides are self-contained, but the material is expanded in a textbook:



The book is available for free as a PDF or via Amazon for a printed copy:

`https://sscardapane.it/alice-book/`

The book was recently published – would be happy for feedback or ideas for completed exercises or additional sections!

Additional useful textbooks:

- **Dive into Deep Learning**, online, much more practical.
- **Understanding Deep Learning**, high-quality illustrations and descriptions.
- **Patterns, Predictions, and Actions**, slightly broader on the machine learning side.
- **Deep Learning - Foundations and Concepts**, for a beginner Bayesian treatment.

From mobile, you can also check out **The Little Book of Deep Learning**.